

Smart Energy-Aware Home Automation System

Project Report

Embedded Systems / IoT Mini Project

1. Introduction

Home automation is one of the fastest-growing applications of the Internet of Things (IoT). Smart home systems provide convenience, security, and energy efficiency by allowing users to monitor and control electrical devices remotely. Traditional electrical switches require manual operation and offer no insight into how much power an appliance is consuming, nor any protection if a fault causes excessive current draw.

Traditional home automation systems only provide switching functionality. However, modern systems combine automation with energy monitoring and protection features. The Smart Energy-Aware Home Automation System integrates appliance control with current monitoring and overload protection to create a safer and smarter home environment, addressing the limitations of conventional installations in a single, low-cost platform built around the ESP32 microcontroller.

This project presents a system that allows users to control up to four electrical appliances wirelessly using an ESP32 microcontroller through a Wi-Fi-based web interface. The system integrates a 4-channel relay module for switching appliances ON and OFF, an ACS712 current sensor for real-time current monitoring, and a 16x2 I2C LCD display for showing appliance status and power consumption. An overload detection mechanism is also incorporated, which automatically disconnects all appliances when excessive current is detected, thereby enhancing safety and preventing equipment damage.

1.1 Background

As electricity costs rise and concerns about energy conservation grow, there is increasing value in systems that can both control and report on appliance usage. The ESP32 microcontroller is chosen for this project due to its built-in Wi-Fi capability, low power consumption, and high processing performance, making it well suited for retrofitting existing electrical systems with smart capabilities. The ACS712 sensor enables real-time current monitoring, while the relay module allows control of multiple appliances, and the I2C LCD provides immediate local feedback regarding system status and power consumption.

1.2 Problem Statement

Conventional switchboards offer no insight into appliance-level power consumption and no automatic protection against overcurrent conditions caused by faults or overloading. Users must rely on manual observation and circuit breakers, which react only after a fault has already occurred, often with some delay. There is a need for a low-cost system that can both control appliances remotely and proactively monitor current draw to prevent damage before it happens.

1.3 Scope of the Project

This project focuses on a four-channel prototype suitable for controlling small household appliances such as lamps, fans, and similar single-phase loads. The scope includes Wi-Fi based wireless ON/OFF control through a web interface, real-time current measurement and display, and automatic overload cutoff for all connected loads. Advanced features such as mobile applications, voice control, and cloud connectivity are identified as potential future extensions but are not implemented in the current prototype.

2. Objectives

The main objectives of this project are outlined below:

- To develop an IoT-based smart home automation system using the ESP32 microcontroller.
- To remotely control up to four household appliances using a Wi-Fi based web interface.
- To monitor real-time current consumption of connected loads using the ACS712 current sensor.
- To display appliance status and current values in real time on a 16x2 I2C LCD.
- To implement automatic overload detection and protection that disconnects all appliances when a fault condition is detected.
- To improve overall energy efficiency and electrical safety in household environments.
- To provide a scalable, low-cost platform that can be extended with additional sensors, appliances, or connectivity features in the future.

Each of these objectives addresses a specific gap in conventional switching systems. Wireless control removes the need for physical proximity to operate appliances, allowing the user to manage devices from a smartphone or computer connected to the same Wi-Fi network. Real-time current monitoring provides visibility into energy consumption patterns that are otherwise invisible to the user, helping to identify faulty equipment and reduce wastage.

The LCD display ensures that key system information is accessible locally at all times, even without network access. Finally, the overload protection mechanism acts as an automated safeguard, disconnecting all relays and warning the user before excessive current can cause damage to appliances, wiring, or pose a fire hazard.

3. Technology Used

The system is built using a combination of hardware components and software tools, each selected for its suitability to the project's requirements of low cost, ease of interfacing, and reliable performance. The major components used in the system are summarized in the table below.

Component	Purpose / Quantity
ESP32 Development Board	Central controller with built-in Wi-Fi (1)
4-Channel Relay Module	Switches up to four appliances ON/OFF (1)
ACS712 Current Sensor	Measures real-time load current (1)
16x2 I2C LCD Display	Displays appliance status and current values (1)
Jumper Wires	Circuit connections (as required)
USB Cable	Programming and power supply (1)
Power Supply	Powers the system and connected loads (1)
Electrical Appliances	Loads controlled and monitored (up to 4)

3.1 ESP32 Development Board

The ESP32 serves as the central processing and communication unit of the system. It is a powerful, low-cost microcontroller with built-in Wi-Fi and Bluetooth connectivity, multiple GPIO pins, and an analog-to-digital converter (ADC). After connecting to a Wi-Fi network, the ESP32 hosts a web server through which users can send ON/OFF commands to control connected appliances.

3.2 4-Channel Relay Module

The 4-channel relay module is used to switch the connected appliances ON and OFF. Each relay channel acts as an electrically controlled switch, allowing the low-voltage ESP32 GPIO pins to safely control higher-power AC appliances. The module provides electrical isolation between the control circuit and the load circuit, which is essential for user safety.

3.3 ACS712 Current Sensor

The ACS712 is a Hall-effect based linear current sensor that provides an analog output voltage proportional to the current flowing through it. In this project, it is used to continuously measure the current drawn by the connected appliances, enabling real-time monitoring of power consumption and forming the basis of the overload detection feature.

3.4 16x2 I2C LCD Display

The 16x2 I2C LCD display provides a local visual interface for the system. Using the I2C protocol significantly reduces the number of wires required compared to a standard parallel LCD

interface, needing only two data lines (SDA and SCL) in addition to power and ground. The display shows appliance status and real-time current readings, and presents an overload warning message when triggered.

3.5 Software Tools

The system firmware is developed using the Arduino IDE, which provides a straightforward environment for writing, compiling, and uploading code to the ESP32. The Arduino ecosystem offers libraries for hosting a web server on the ESP32, interfacing with the I2C LCD, and reading analog sensor values, which together simplify the development process significantly.

4. Methodology

The system is built around the ESP32 microcontroller, which acts as the central processing and communication unit. The design and implementation methodology followed for this project is described in the sections below, covering circuit connections, working principle, and the control algorithm.

4.1 Circuit Connections

The hardware components are interconnected with the ESP32 as summarized in the tables below.

Relay Module Connections:

Relay Module Pin	ESP32 Pin
VCC	5V
GND	GND
IN1	GPIO 26
IN2	GPIO 27
IN3	GPIO 14
IN4	GPIO 12

ACS712 Current Sensor Connections:

ACS712 Pin	ESP32 Pin
VCC	5V
GND	GND
OUT	GPIO 34

I2C LCD Connections:

LCD Pin	ESP32 Pin
VCC	5V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

4.2 Working Principle

The ESP32 acts as the central controller of the system. After connecting to a Wi-Fi network, it hosts a web server through which users can control electrical appliances from any device on the same network.

Each appliance is connected to one relay channel. When a user sends a command through the web interface, the ESP32 receives the request and activates or deactivates the corresponding relay, thereby switching the appliance ON or OFF.

In parallel, the ACS712 current sensor continuously measures the current drawn by the connected appliances. The measured analog value is read by the ESP32's ADC, converted into a current value using calibration based on the sensor's sensitivity, and displayed in real time on the I2C LCD.

If the measured current exceeds a predefined threshold, the ESP32 automatically switches off all relays and displays an overload warning message on the LCD. This protects the appliances and wiring from damage caused by excessive current, and the system resumes normal operation once the fault condition is cleared and the user issues a fresh command.

4.3 System Flow

The overall data and control flow of the system can be summarized as follows: the user sends ON/OFF commands through a web browser; the ESP32 processes these commands and drives the relay module to switch appliances accordingly, while the ACS712 sensor continuously measures current and the LCD displays appliance status and readings. Overload conditions detected at any point trigger the automatic protection routine, overriding any active relay states.

4.4 Control Algorithm

The firmware logic implemented on the ESP32 follows the algorithm below:

1. Start the system.
2. Initialize the ESP32, LCD, Wi-Fi connection, relay outputs, and current sensor.
3. Connect the ESP32 to the configured Wi-Fi network and start the web server.
4. Wait for user commands from the web interface.
5. Read the current value from the ACS712 sensor.
6. Display the current value and appliance status on the LCD.
7. If a user sends an ON/OFF command, activate the corresponding relay and update the appliance status.
8. Check the measured current against the overload threshold.
9. If the current exceeds the threshold, turn OFF all relays and display an overload warning on the LCD.
10. Repeat the process continuously.

4.5 Hardware Design Considerations

Electrical isolation between the ESP32's low-voltage control circuit and the relay outputs is essential for safety, provided by the relay module's opto-isolated inputs. Wiring for the ACS712 sensor was kept in series with the live conductor of the load circuit to ensure accurate current

measurement, with care taken to avoid loose connections that could introduce measurement noise. The system was designed with modularity in mind, allowing each relay channel to be tested independently and the system to be scaled to additional channels with minimal changes.

5. Output / Result

The implemented system was tested to verify each of its core functions, with the following results observed.

5.1 Wireless Appliance Control

The system successfully allows wireless control of up to four electrical appliances through the ESP32's Wi-Fi based web interface. The 4-channel relay module responded correctly to ON/OFF commands sent from a web browser, switching the connected loads as required without noticeable delay.

5.2 Real-Time Current Monitoring

The ACS712 current sensor accurately measured the current consumption of the connected appliances, and these readings were displayed in real time on the 16x2 I2C LCD along with the corresponding appliance status. The measured values were found to be consistent with expected current draws for the test loads used.

5.3 Overload Protection

During testing, when the total current exceeded the set threshold, the system automatically turned OFF all relays and displayed an overload warning message on the LCD, demonstrating effective overload protection. This confirms that the system meets its objectives of providing wireless control, real-time monitoring, and automatic safety protection.

5.4 Summary of Results

- All four relay channels operated correctly under wireless control commands sent via the web interface.
- Current readings displayed on the LCD matched expected values for connected loads.
- Overload condition was detected and all relays were switched off automatically within the expected response time, with a warning message shown on the LCD.
- The system operated reliably over repeated test cycles without unexpected resets or communication failures, confirming stable Wi-Fi connectivity and low power consumption during operation.

6. Conclusion

The Smart Energy-Aware Home Automation System successfully combines appliance control, energy monitoring, and safety features into a single, cost-effective platform. The ESP32 provides wireless connectivity and hosts the control interface, the relay module switches the connected appliances, the ACS712 sensor monitors current consumption, and the I2C LCD displays system information in real time.

By combining ESP32-based wireless control, real-time current monitoring using the ACS712 sensor, and an automatic overload protection mechanism, the system enhances both convenience and safety in everyday appliance usage. The real-time display of appliance status and power consumption on the I2C LCD further improves user awareness of energy usage, allowing users to identify high-consumption appliances and make informed decisions to reduce energy wastage.

The overload protection feature adds an important layer of safety, automatically disconnecting all appliances before excessive current can cause damage or pose a fire hazard. This project demonstrates the practical implementation of IoT technology in modern smart homes and contributes to efficient energy management and enhanced user convenience.

Overall, the project meets all of its stated objectives and provides a solid foundation for a functional smart home automation solution. This system lays a strong foundation for future enhancements such as integration with mobile applications, voice control using Google Assistant or Alexa, cloud-based energy monitoring and analytics, automatic scheduling of appliances, SMS and email alert systems, and AI-based smart energy optimization.